

## Medtronic Hugo Surgical Robot – Hardware and Software

Senior Mechanical Engineer, Senior Controls Software Engineer, 2022-2026

### Mechanical Subsystem Tech Lead

Led design and delivery of Control Arm subsystems (actuator, pin brake, emergency release) from concept through mechanical validation using Creo, combining mechanical design (FEA, tolerances, ASME Y14.5), manufacturing optimization (CNC/molding/sheet metal), and IEC 60601 compliance

### Control Systems & Software

Architected Simulink control software for 6-DOF surgical arm; delivered 37 functional modules with 77 verification test cases; redesigned plug-and-play instrument controller

### Field Troubleshooting & Production Validation

Investigated and resolved field failures through root cause analysis and vendor testing; validated designs on international user studies; released production improvements across deployed systems

### Design for Manufacturing

Optimized designs across CNC, injection molding, sheet metal, and 3D printing; collaborated with suppliers on manufacturability validation and process improvement

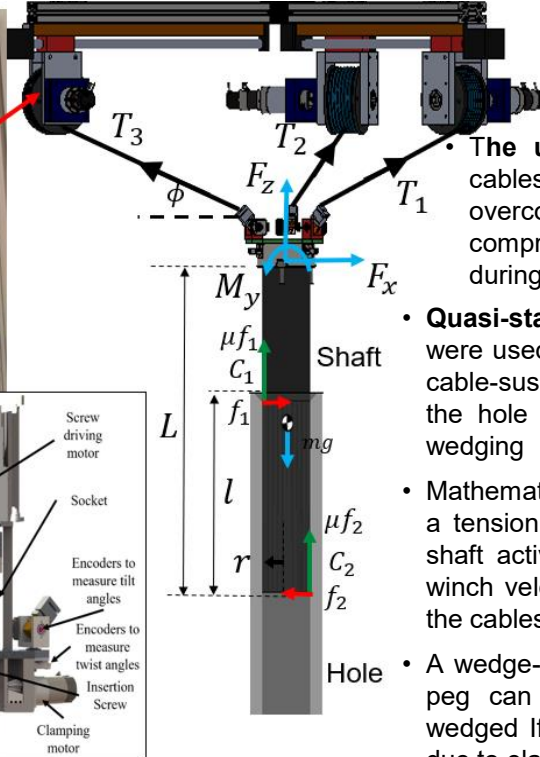
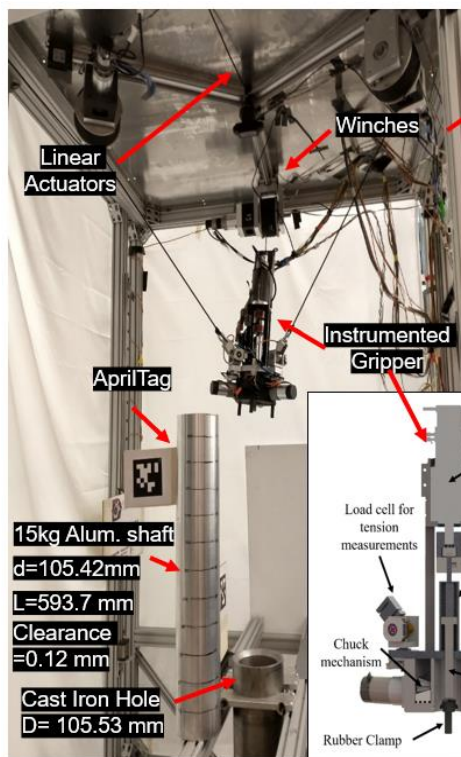
### Technical Leadership & Expert

Robotic Arm hardware/software SME; lead cross-functional design decisions; expert in GD&T, IEC 60601, IEC 62304, ISO 1497, ISO 14971



## Precision Assembly of Suspended Heavy Objects: Winchbot

PhD Research, 2019-2021



• The WinchBot system aims to revolutionize the manufacturing industry by automating the grasping and insertion of heavy shafts into gearboxes.

• The uni-directional load bearing properties of cables were exploited so that the shaft is not overconstrained. The cables go slack if under compressive load, **reducing damage** to the parts during assembly

• Quasi-static force balance and kinematic analysis were used to determine the conditions required for a cable-suspended shaft to cross the chamfer, enter the hole and be fully inserted without jamming or wedging

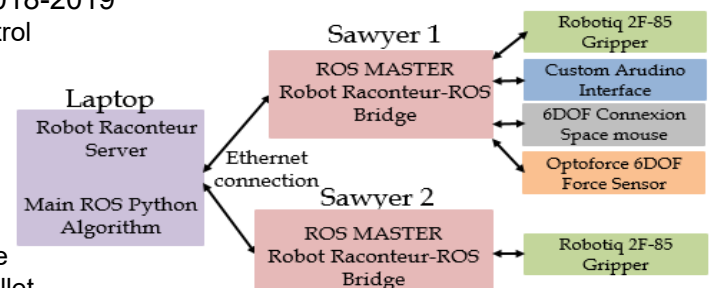
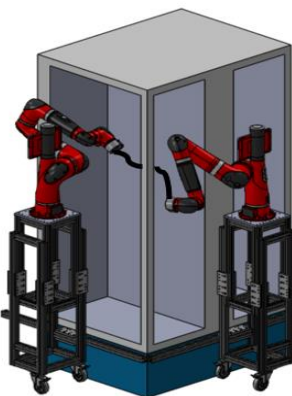
• Mathematical system modeling was used to develop a tension balancing feedback controller so that the shaft actively avoids jamming during insertion. The winch velocities are regulated so that the tension in the cables remain balanced.

• A wedge-breaking model was developed so that the peg can be reinserted in the event it becomes wedged. If the peg becomes wedged within the hole due to elastic deformation.

## Robot Assisted Wire Harness Installation

PhD Research, 2018-2019

- Developed Teach-By-Demonstration a control strategy coordinating a dual-arm robotic system for positioning wire harness while avoiding collisions
- Implemented Dual-Arm Impedance and Admittance Control for safe human-robot interaction using python and ROS
- Developed a custom path planning software using Gazebo, two sawyer robots and pybullet physics simulator for collision detection

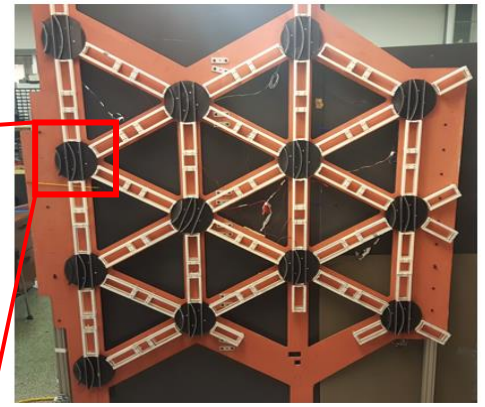
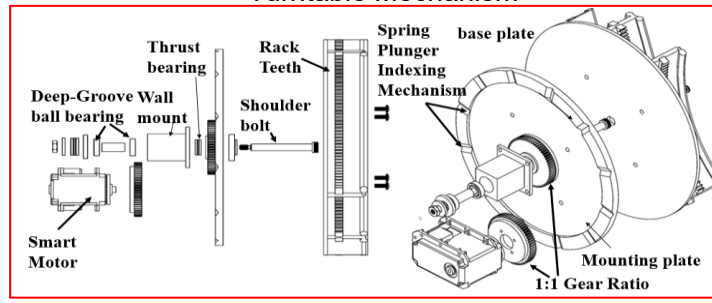


## A Multi-Track Elevator System for E-Commerce Fulfillment Centers

Master's Research 2015-2017

Designed and fabricated a novel multi-track elevator system that uses a rack-and-pinion mechanism to drive carriages and a turntable facilitating direction changes

### Turntable Mechanism

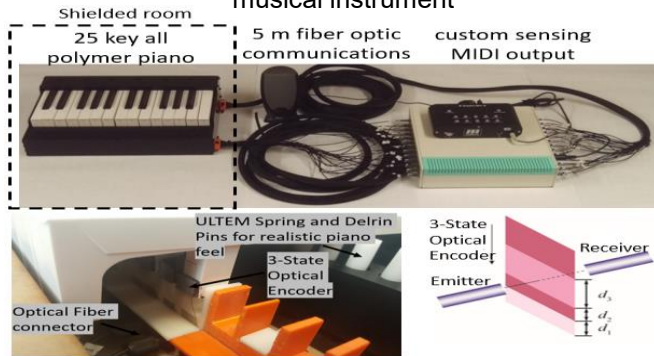


### As a student

## Medical Device Design Course

### As a mentor

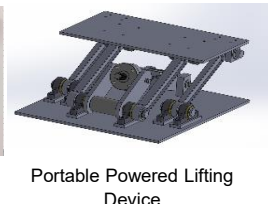
Designed a Magnetoencephalography (MEG) compatible musical instrument



Lower Extremity Fracture Fixation Device



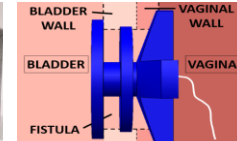
Crank Powered Otoloscope



Portable Powered Lifting Device



Center of Gravity Detector for Operator Fatigue



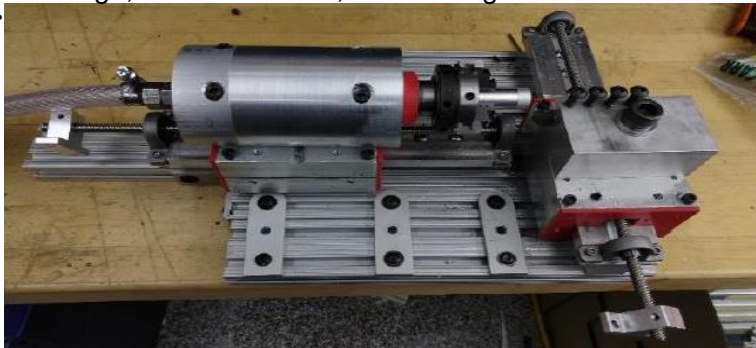
Vesicovaginal Fistula Occluder



Vaginal Breech Delivery Device

## Precision Machine Design Course 2018

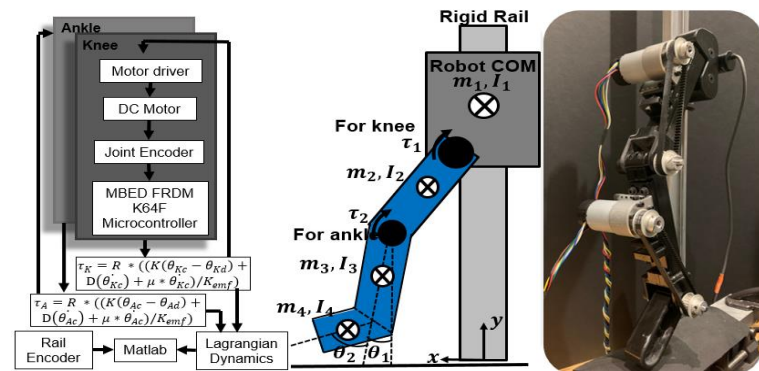
Design, Error Prediction, and Testing of a T-Based Lathe



- Measured flatness of machined aluminum: 187.9  $\mu\text{m}$   $\pm$  5.75  $\mu\text{m}$
- Error Budget Predicted flatness 233  $\mu\text{m}$   $\pm$  22.7  $\mu\text{m}$
- Custom spindle and custom boxway linear motion axes mounted perpendicular to one-another

## Bio Inspired Robotics Course 2019

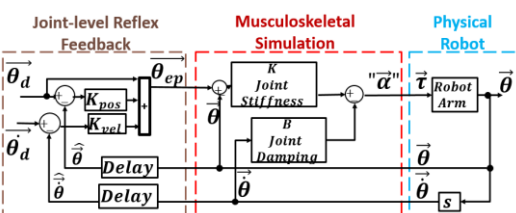
Study to determine if active impedance and angle of attack can minimize COM jerk



## Realistic Human Reaching Movements in a Direct-Drive Robot Arm using Virtual Muscle Impedance and Equilibrium Point Control

Biomechanics and Neural Control of Movement Course 2016

Stiffness + Reflex Feedback Controller from McIntyre and Bizzzi's servo hypothesis and Flash's lumped-parameter multi-joint equilibrium stiffness model



90 Watt Maxon EC90 flat brushless motors  
 Ø90 mm with Hall sensors

## Disinfection Robot (2020):

Developed a mobile platform that can be used to disinfect a room with UV light. This platform is used for teaching introduction to robotics.

